

Personal Budget Manager - Execution Documentation

Course: Advanced Python Programming

Project: Personal Budget Manager

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1. Application Menus & Navigation

1.1 Main Application Menu

The entry point of the system displaying the primary options for navigation.

```
=====
💰  PERSONAL BUDGET MANAGER - MAIN MENU  📁
=====
[1] 📄 Transaction Management
[2] 🎯 Monthly Budget & Alerts
[3] 📊 Financial Reports & Analysis
[4] 🎨 Data Visualization & Charts
[5] 💾 Save & Exit
=====
Select an option (1-5):
```

Figure 1.1: Main Application Menu interface.

1.2 Transaction Management Sub-Menu

The interactive sub-menu dedicated to transaction operations (Add, View, Edit, Delete, Search).

```
Select an option (1-5): 1
-----
📄 TRANSACTION MANAGEMENT
-----
[1] 📄 View All Transactions
[2] ➕ Add New Transaction
[3] ✎ Edit Existing Transaction by ID
[4] ✖ Delete Transaction by ID
[5] 🔍 Search & Filter Transactions
[6] ⬅ BACK Back to Main Menu
```

Figure 1.2: Transaction Management sub-menu interface.

2. Transaction Management & Data Persistence

Demonstrating data loading from `transactions.csv` and displaying all stored Income and Expense records.

```
Select an option (1-6): 1
--- Total Transactions ---
- 1,2026-06-01,income,Salary,June Salary,12000.0,Company
- 2,2026-06-02,expense,Food,Supermarket Groceries,450.0,Shufersal,Credit Card
- 3,2026-06-03,expense,Transport,Monthly Bus Pass,210.0,Rav Kav,Cash
- 4,2026-06-05,income,Freelance,Web Design Project,2500.0,Client
- 5,2026-06-10,expense,Food,Restaurant Dinner,180.0,Japanika,Credit Card
- 6,2026-06-15,expense,Entertainment,Cinema Tickets,90.0,Planet Cinema,Credit Card
- 7,2026-06-20,expense,Bills,Electricity Bill,320.0,Electric Co,Bank Transfer
- 8,2026-07-01,income,Salary,July Salary,12000.0,Company
```

Figure 2: Viewing loaded transactions from CSV in terminal.

3. Financial Reporting & Analytics (Pandas)

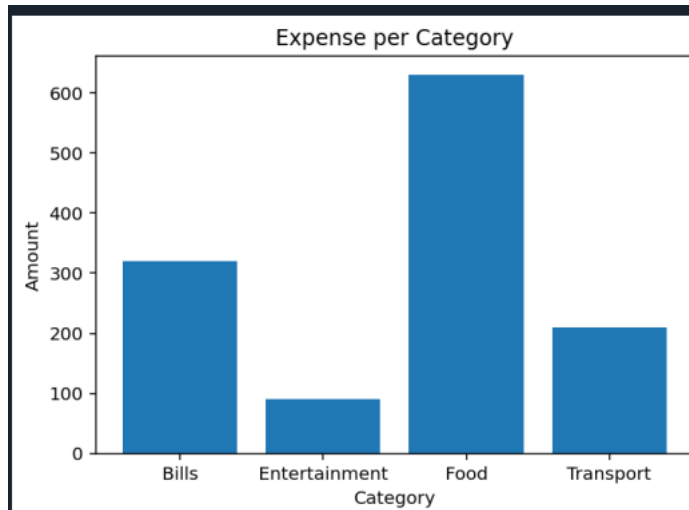
Demonstrating automated financial analysis using Pandas DataFrames (Financial Overview / Monthly Summary).

```
Select an option (1-9): 4
--- MONTHLY SUMMARY ---
Year_Month  expense  income
0    2026-06    1250.0  14500.0
1    2026-07         0.0  12000.0
```

Figure 3: Financial Overview report generated using Pandas.

4. Data Visualization (Matplotlib Chart)

Demonstrating data visualization capabilities by generating a Category Expense Bar Chart.



Select an option (1-3): 1

--- EXPENSE BY CATEGORY CHART ---

Important

Figures are displayed in the Plots pane by default. To make them also appear inline in the console, you need to uncheck "Mute inline plotting" under the options menu of Plots.

✓ Chart saved successfully as 'expenses_by_category.png'

Figure 4: Expense by Category visualization created with Matplotlib.